

Experiment 3

Module Name: Robot Framework

Experiment Name: Custom Keywords and Libraries

Problem Statement: Build a custom Robot Framework keyword using Python to perform a specific action, such as calculating the sum of two numbers. Utilize the "BuiltIn" library to manipulate strings or perform mathematical operations within your test cases

Objective:. To extend Robot Framework's capabilities by writing custom Python libraries and leveraging core BuiltIn functions.

Tools, Software, and Concepts Used:

Python Functions (.py files), Library Importing, BuiltIn library keywords (Evaluate)

Source Code / Implementation Steps:

Step 1: Create a Python file named CustomMath.py

```
def calculate_sum(num1, num2):  
    return float(num1) + float(num2)
```

Step 2: Create a Robot file 'Experiment3.robot' and write this:

```
*** Settings ***  
Library      CustomMath.py  
Library      BuiltIn  
  
*** Test Cases ***  
Test Custom Python Keyword  
    [Documentation]    Calls the custom python calculation keyword.  
    ${result}=        Calculate Sum      15.5      4.5  
    Log              The custom sum is: ${result}  
  
Test BuiltIn Math Operation  
    [Documentation]    Uses BuiltIn Evaluate for math.  
    ${builtin_result}=    Evaluate      15.5 + 4.5  
    Log              The builtin sum is: ${builtin_result}
```

Step 3: Now from the folder go to cmd and write this:

```
>>>python -m robot Experiment3.robot
```

Output with clearly labelled screenshots:

```
D:\My Desktop\Robot Project>python -m robot Experiment3.robot
=====
Experiment3
=====
Test Custom Python Keyword :: Calls the custom python calculation ... | PASS |
-----
Test BuiltIn Math Operation :: Uses BuiltIn Evaluate for math.          | PASS |
-----
Experiment3                                                             | PASS |
2 tests, 2 passed, 0 failed
=====
Output: D:\My Desktop\Robot Project\output.xml
Log:     D:\My Desktop\Robot Project\log.html
Report:  D:\My Desktop\Robot Project\report.html
```

Screenshot 1: Terminal Output

Experiment3 Log

Generated: 20260921 00:10:35 UTC+05:30
2 minutes 21 seconds ago

Statistics

Total Statistics	Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
All Tests	2	2	0	0	00:00:00	

Statistics by Tag

Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
No Tags					

Statistics by Suite

Total	Pass	Fail	Skip	Elapsed	Pass / Fail / Skip
Experiment3	2	2	0	0	00:00:00

Execution Log

Suite	Test Name	Full Name	Source	Start / End / Elapsed	Status	Time
Experiment3	Test Custom Python Keyword	Experiment3	D:\My Desktop\Robot Project\Experiment3.robot	20260921 00:10:35.372 / 20260921 00:10:35.448 / 00:00:00.076	2 tests total, 2 passed, 0 failed, 0 skipped	00:00:00.076
	Test Custom Python Keyword					00:00:00.004
	Test BuiltIn Math Operation					00:00:00.002

Screenshot 2: Output of log.html

Brief Result: Both the custom Python script and the BuiltIn Evaluate keyword successfully added the numbers and stored the results.

Observation: Custom Python functions are automatically mapped to Robot Framework keywords (e.g., calculate_sum becomes Calculate Sum).

Conclusion: The framework is highly extensible. While BuiltIn handles most standard operations, custom Python libraries allow for infinite integration possibilities with internal business logic.